

Real functions of a real variable



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Real function of a real variable

Introduction: Real function of a real variable

- ▶ In this section we focus on functions whose input and output are **real numbers**
- ▶ We introduce the **graph** of a function, the main tool to visualize how $f(x)$ depends on x
- ▶ We will see that **not all curves in the plane** are graphs of functions

Real function of a real variable

Definition 1: real function of a real variable

A function whose domain D is a subset of $\mathbb{R}$ and whose codomain is $\mathbb{R}$:

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto f(x)$$

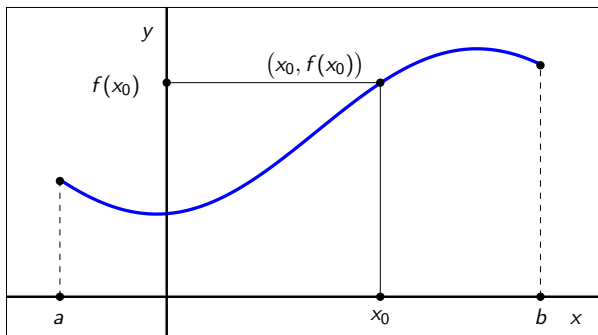
*is called a **real function of a real variable**.*

- ▶ These are functions in which the “input” variable x and the “output” variable $f(x)$ are real numbers.
- ▶ The most common real functions of a real variable have as domain D and as image $f(D)$ an **interval** (possibly the whole of $\mathbb{R}$) or the union of a **finite number of intervals**.

The graph of a function (1/3)

The dependence of the output $f(x)$ on the input x is effectively visualized by drawing the **graph** of f , that is, the set of points of the plane with coordinates (x, y) such that $y = f(x)$ and x in the domain D .

Example of the graph of a real function of a real variable with domain $D = [a, b]$ (a closed and bounded interval):



The graph of a function (2/3)

- ▶ Every line parallel to the y -axis that intersects the x -axis at a point x_0 of the domain D intersects the graph of f at one and only one point.
- ▶ If the line did not intersect the graph, this would mean that no output corresponds to the input x . If the line intersected the graph at more than one point, this would mean that several distinct outputs correspond to the input x , and hence the function would no longer be single-valued.

Therefore, not all curves are graphs of functions.

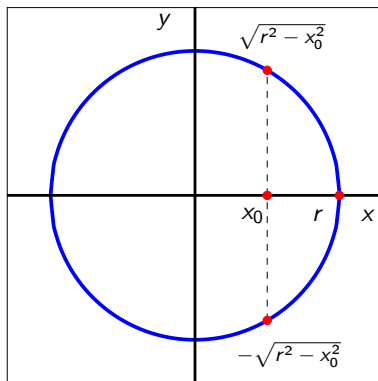
The graph of a function (3/3)

- Note, instead, that nothing prevents a line parallel to the x -axis from intersecting the graph of f at several points or at no point at all.

Example (1/2): curve that does not correspond to the graph of a function

Consider, for example, the curve made of the points of the circle of radius r :

$$x^2 + y^2 = r^2 \iff y = \pm \sqrt{r^2 - x^2}$$



Example (2/2): curve that does not correspond to the graph of a function

- ▶ The vertical line $x = x_0$ intersects the circle at the two points with ordinates $\sqrt{r^2 - x_0^2}$ and $-\sqrt{r^2 - x_0^2}$.

It is not possible to associate a unique output with the input $x_0 \in (-r, r)$, hence this curve does not correspond to the graph of a function!

Real functions of a real variable:

- ▶ $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$: input and output are real numbers
- ▶ The **graph** of f is the set of points $(x, f(x))$ with $x \in D$
- ▶ Every vertical line through a point of the domain intersects the graph at **one and only one point**
- ▶ Not all curves are graphs of functions (e.g. the circle)

Sign of a function

Introduction: Sign of a function

- ▶ The properties of a function are almost always studied not only on the whole domain, but also on a **part** of it
- ▶ One says for instance that $x \mapsto x^2$ is “positive for $x > 0$ ” or “decreasing in $(-\infty, 0]$ ”
- ▶ We therefore state the definitions directly on a subset I of the domain, and recover the “global” property as the particular case $I = D$

Sign of a function

Definition 2: sign of a function in a subset of the domain

Given a function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and a subset $I \subseteq D$, the function is said to be

$$\left\{ \begin{array}{ll} \text{nonnegative in } I & \text{if } f(x) \geq 0, \forall x \in I \\ \text{positive in } I & \text{if } f(x) > 0, \forall x \in I \\ \text{nonpositive in } I & \text{if } f(x) \leq 0, \forall x \in I \\ \text{negative in } I & \text{if } f(x) < 0, \forall x \in I \end{array} \right.$$

From local to global

- ▶ When the subset is not specified, the **whole domain** is understood: f is called *nonnegative* (*positive*, *nonpositive*, *negative*) if it is so in $I = D$
- ▶ Graphically: f is nonnegative in I if no point of the graph with abscissa in I lies *below* the x -axis; it is positive in I if all these points lie *strictly above* it
- ▶ If f is positive in I then it is also nonnegative in I ; the converse is false, because f may vanish at some point of I

Example: sign of a function

- ▶ $f : \mathbb{R} \rightarrow \mathbb{R}$, $f : x \mapsto x^2$ is *nonnegative* in $\mathbb{R}$, since $x^2 \geq 0$ for every x . It is not *positive* in $\mathbb{R}$, because $f(0) = 0$: it is positive in $(0, +\infty)$ and in $(-\infty, 0)$
- ▶ $f : \mathbb{R} \rightarrow \mathbb{R}$, $f : x \mapsto x^3$ is *positive* in $(0, +\infty)$ and *negative* in $(-\infty, 0)$:

$$x^3 > 0 \iff x > 0 \quad \text{and} \quad x^3 < 0 \iff x < 0$$

- ▶ $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f : x \mapsto \frac{1}{x}$ is *positive* in $(0, +\infty)$ and *negative* in $(-\infty, 0)$

Summary

Sign of a function:

- ▶ The properties are stated on a subset $I \subseteq D$ of the domain; with no indication, $I = D$ is understood
- ▶ **Nonnegative** in I : $f(x) \geq 0$ for every $x \in I$; **positive** in I : $f(x) > 0$
- ▶ **Nonpositive** in I : $f(x) \leq 0$ for every $x \in I$; **negative** in I : $f(x) < 0$
- ▶ Positive $\Rightarrow$ nonnegative; the converse does not hold (e.g. x^2 in $\mathbb{R}$)

Bounded functions



Introduction: Bounded functions

- ▶ In this section we study when the values of a function stay **confined** below, above or between two values
- ▶ We introduce functions that are **bounded above**, **bounded below** and **bounded** in a subset I of the domain, with their graphical interpretation
- ▶ These concepts will be fundamental in the study of limits and of maxima and minima

Bounded functions

Definition 3: bounded functions in a subset of the domain

Given a function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and a subset $I \subseteq D$, the function is said to be

$$\left\{ \begin{array}{ll} \text{bounded above in } I & \text{if } \exists M \in \mathbb{R} : f(x) \leq M, \quad \forall x \in I \\ \text{bounded below in } I & \text{if } \exists m \in \mathbb{R} : f(x) \geq m, \quad \forall x \in I \\ \text{bounded in } I & \text{if } \exists M \in \mathbb{R}_{\geq 0} : |f(x)| \leq M, \quad \forall x \in I \end{array} \right.$$

Upper and lower bounds

- ▶ A number $M \in \mathbb{R}$ such that $f(x) \leq M$ for every $x \in I$ is called an **upper bound** for the values of f in I
- ▶ A number $m \in \mathbb{R}$ such that $f(x) \geq m$ for every $x \in I$ is called a **lower bound** for the values of f in I
- ▶ Hence f is bounded above in I if its values in I admit an upper bound, and bounded below in I if they admit a lower bound
- ▶ When the subset is not specified, the **whole domain** is understood ($I = D$)

Bounded = bounded above and below

A function is bounded in I *if and only if* it is bounded both above and below in I .

- ▶ If $|f(x)| \leq M$ for every $x \in I$, then $-M \leq f(x) \leq M$: hence M is an upper bound and $-M$ a lower bound
- ▶ Conversely, if $m \leq f(x) \leq M$ for every $x \in I$, setting $K = \max\{|m|, |M|\} \geq 0$:

$$-K \leq m \leq f(x) \leq M \leq K \quad \text{that is} \quad |f(x)| \leq K$$



Graphical interpretation

► Graphically, we have:

1. a function is bounded above if its graph is contained in the lower half-plane bounded by a line parallel to the x -axis
2. a function is bounded below if its graph is contained in the upper half-plane bounded by a line parallel to the x -axis
3. a function is bounded if its graph is contained in a horizontal strip

Example (1/2): bounded function

Consider the function:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \frac{1}{1+x^2} + 1$$

we have

$$\frac{1}{1+x^2} + 1 = \frac{1+x^2-x^2}{1+x^2} + 1 = 2 - \frac{x^2}{1+x^2}$$

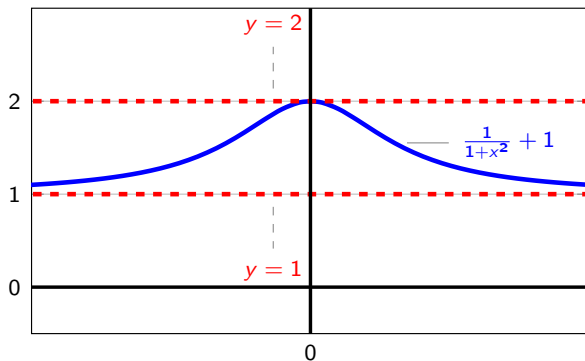
and

$$\frac{x^2}{1+x^2} \geq 0, \quad \forall x \in \mathbb{R}$$

hence

$$1 < \frac{1}{1+x^2} + 1 \leq 2, \quad \forall x \in \mathbb{R}$$

Example (2/2): bounded function

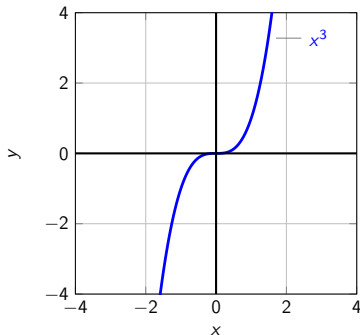


Example: unbounded function

The function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^3 \quad (y = x^3)$$

is bounded neither above nor below.

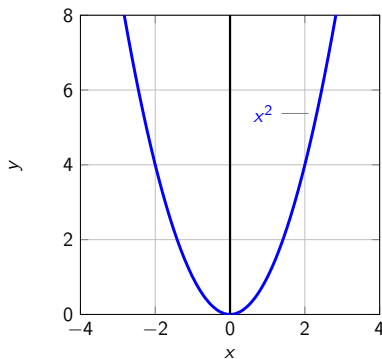


Example: function bounded below

The function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^2 \quad (y = x^2)$$

is bounded below; indeed, $x^2 \geq 0, \forall x \in \mathbb{R}$.



Example: boundedness in a subset of the domain

A function may fail to be bounded on its domain and still be bounded on a subset of it.

- ▶ $f : \mathbb{R} \rightarrow \mathbb{R}$, $f : x \mapsto x^3$ is not bounded in $\mathbb{R}$, but it is bounded in $I = [-2, 2]$: for $-2 \leq x \leq 2$

$$|x^3| = |x|^3 \leq 2^3 = 8$$

- ▶ $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f : x \mapsto \frac{1}{x}$ is **not** bounded above in $I = (0, +\infty)$: given $M > 0$, choosing $0 < x < \frac{1}{M}$ we get $f(x) = \frac{1}{x} > M$. It is instead bounded in $I = [1, +\infty)$, where $0 < \frac{1}{x} \leq 1$

Boundedness and image

- Equivalently, we can say that a function is *bounded above* (*bounded below*, *bounded*) if, respectively, its **image** is a subset of $\mathbb{R}$ that is *bounded above* (*bounded below*, *bounded*).

Summary

Bounded functions:

- ▶ **Bounded above** in I : $\exists M : f(x) \leq M$ for every $x \in I$ (M upper bound)
- ▶ **Bounded below** in I : $\exists m : f(x) \geq m$ for every $x \in I$ (m lower bound)
- ▶ **Bounded** in I : $\exists M \geq 0 : |f(x)| \leq M$ for every $x \in I$ (graph in a horizontal strip)
- ▶ Examples: $\frac{1}{1+x^2} + 1$ is bounded, x^3 is unbounded in $\mathbb{R}$ but bounded in $[-2, 2]$, x^2 is bounded below

Symmetric functions



Introduction: Symmetric functions

- ▶ In this section we study functions whose graph has a **symmetry**
- ▶ We introduce **even** functions (symmetry with respect to the y -axis) and **odd** functions (symmetry with respect to the origin)
- ▶ Recognizing a symmetry halves the work when studying a function

Definition 4: even function

*Functions whose graph is symmetric with respect to the y-axis are called **even**.*

- ▶ They are characterized by the relation

$$f(-x) = f(x)$$

which expresses the equality of the ordinates corresponding to the points x and $-x$, which are symmetric with respect to $x = 0$.

Definition 5: Odd function

*Functions whose graph is symmetric with respect to the origin are called **odd**.*

- ▶ They are characterized by the relation

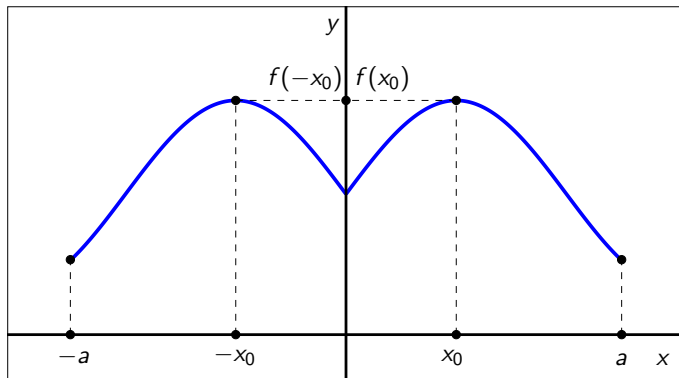
$$f(-x) = -f(x)$$

which expresses the fact that the ordinates corresponding to the points x and $-x$, which are symmetric with respect to $x = 0$, are the opposite of each other.

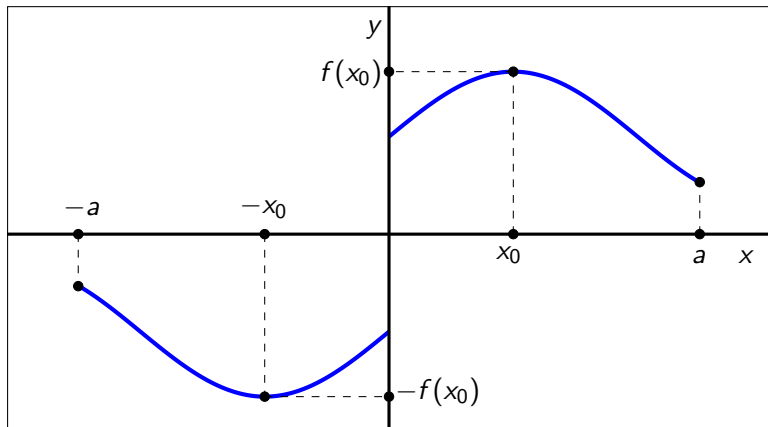
Example: even and odd functions

- ▶ For example, the function $x \mapsto x^2$ is even, while $x \mapsto x^3$ is odd.
- ▶ More generally, powers with integer exponent are even functions if the exponent is even and odd functions if the exponent is odd.

Example of the graph of an even function



Example of the graph of an odd function



No symmetry with respect to the x -axis

A function *cannot* have a graph that is symmetric with respect to the x -axis, since the correspondence would no longer be single-valued.

Summary

Symmetric functions:

- ▶ **Even:** $f(-x) = f(x)$, graph symmetric with respect to the y -axis (e.g. x^2)
- ▶ **Odd:** $f(-x) = -f(x)$, graph symmetric with respect to the origin (e.g. x^3)
- ▶ Powers with integer exponent are even or odd like their exponent
- ▶ No function has a graph symmetric with respect to the x -axis

Monotonic functions



Introduction: Monotonic functions

- ▶ In this section we study how the output of a function changes when the input **increases**
- ▶ We introduce **non-decreasing**, **increasing**, **non-increasing** and **decreasing** functions in a subset I of the domain
- ▶ Monotonicity is one of the fundamental properties in the study of functions and of invertibility

Non-decreasing and increasing functions

Definition 6: Non-decreasing function in a subset of the domain

Given $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $I \subseteq D$, the function is called **non-decreasing in I** if for every pair of points $x_1, x_2 \in I$ we have:

$$x_1 < x_2 \implies f(x_1) \leq f(x_2)$$

Definition 7: Increasing function in a subset of the domain

Given $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $I \subseteq D$, the function is called **increasing in I** if for every pair of points $x_1, x_2 \in I$ we have:

$$x_1 < x_2 \implies f(x_1) < f(x_2)$$

Non-increasing and decreasing functions

Definition 8: Non-increasing function in a subset of the domain

Given $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $I \subseteq D$, the function is called **non-increasing in I** if for every pair of points $x_1, x_2 \in I$ we have:

$$x_1 < x_2 \implies f(x_1) \geq f(x_2)$$

Definition 9: Decreasing function in a subset of the domain

Given $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $I \subseteq D$, the function is called **decreasing in I** if for every pair of points $x_1, x_2 \in I$ we have:

$$x_1 < x_2 \implies f(x_1) > f(x_2)$$

Interpretation

- ▶ A function f is *non-decreasing* in I if, as x increases in I , the corresponding ordinate on the graph does not decrease (it either stays the same or increases); it is *increasing* in I if it always increases
- ▶ A function f is *non-increasing* in I if, as x increases in I , the corresponding ordinate on the graph does not increase (it either stays the same or decreases); it is *decreasing* in I if it always decreases

Monotonic and strictly monotonic functions

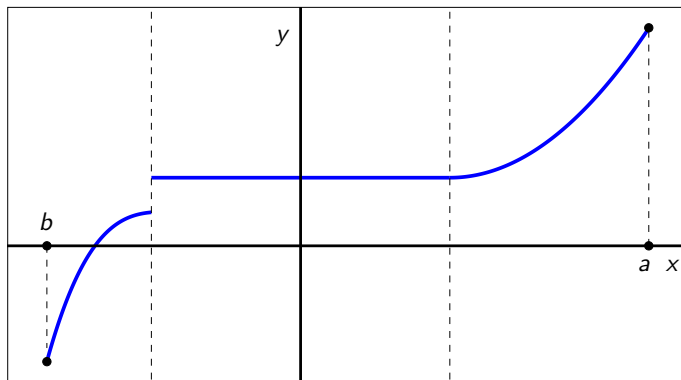
- ▶ A function is called **monotonic in I** if it is *non-decreasing in I* or *non-increasing in I*
- ▶ A function is called **strictly monotonic in I** if it is *increasing in I* or *decreasing in I*
- ▶ When the subset is not specified, the **whole domain** is understood ($I = D$)
- ▶ A function increasing in I is in particular non-decreasing in I , and a decreasing one is non-increasing in I : hence every function **strictly monotonic in I** is also **monotonic in I** ; the converse is false

Example: monotonic functions

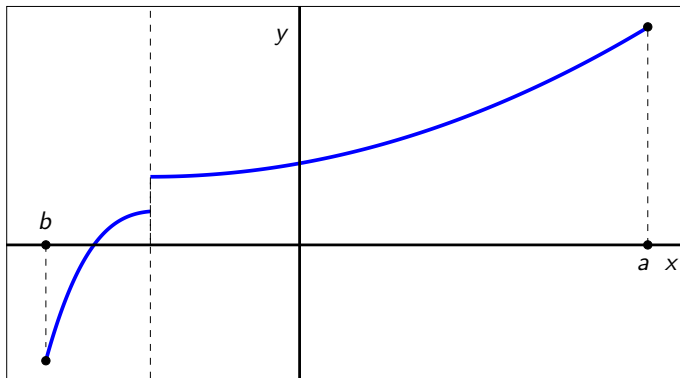
- ▶ The function $x \mapsto x^3$ is *increasing* in $\mathbb{R}$ (hence strictly monotonic in $\mathbb{R}$)
- ▶ The constant function $x \mapsto k$ (whose graph is the line with equation $y = k$) is both *non-decreasing* and *non-increasing* in $\mathbb{R}$: it is monotonic, but it is neither increasing nor decreasing
- ▶ The function $x \mapsto x^2$ is *decreasing* in $(-\infty, 0]$ and *increasing* in $[0, +\infty)$, but it is **not** monotonic in $\mathbb{R}$: indeed $f(-1) = 1 > f(0) = 0$ and $f(0) = 0 < f(1) = 1$

Example of the graph of a non-decreasing function

(horizontal segment)



Example of the graph of an increasing function



- Increasing or decreasing functions (strictly or not) are called **monotonic**.

Monotonic functions:

- ▶ In $I \subseteq D$, with $x_1 < x_2$: **non-decreasing** $f(x_1) \leq f(x_2)$;
increasing $f(x_1) < f(x_2)$
- ▶ In $I \subseteq D$, with $x_1 < x_2$: **non-increasing** $f(x_1) \geq f(x_2)$;
decreasing $f(x_1) > f(x_2)$
- ▶ **Monotonic** in I : non-decreasing or non-increasing in I ;
strictly monotonic in I : increasing or decreasing in I
- ▶ E.g.: x^3 is increasing in $\mathbb{R}$; a constant is monotonic but not strictly; x^2 is decreasing in $(-\infty, 0]$ and increasing in $[0, +\infty)$, but not monotonic in $\mathbb{R}$

Periodic functions

Introduction: Periodic functions

- ▶ In this section we study functions that **repeat themselves** at regular intervals
- ▶ We introduce the **period** and the **periodicity interval**
- ▶ Periodic functions describe cyclic phenomena (oscillations, waves) and include the trigonometric functions

Definition 10: Periodic function

A (non-constant) function $f : D \rightarrow \mathbb{R}$ is **periodic** with period T , $T > 0$, if T is the smallest positive real number such that

$$f(x + T) = f(x) \quad \text{for every } x \in D$$

- ▶ Every interval of length T contained in D is called a **periodicity interval**.

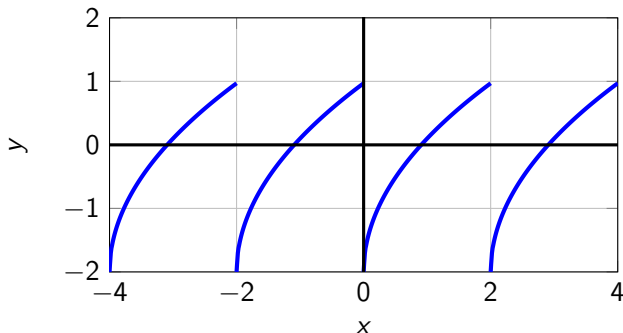
Example: periodic functions

Typical examples of periodic functions are the *trigonometric functions*:

- ▶ $x \mapsto \sin(x)$ ($T = 2\pi$)
- ▶ $x \mapsto \cos(x)$ ($T = 2\pi$)
- ▶ $x \mapsto \tan(x)$ ($T = \pi$)

Example: graph of periodic functions

Graph of a periodic function with period $T = 2$:



Horizontal lines

- ▶ Lines parallel to the x -axis have equation

$$y = k \quad k \in \mathbb{R}$$

Periodic functions:

- ▶ f is **periodic** with period $T > 0$ if T is the smallest positive real number with $f(x + T) = f(x)$ for every $x \in D$
- ▶ Every interval of length T contained in D is a **periodicity interval**
- ▶ Examples: $\sin$ and $\cos$ ($T = 2\pi$), $\tan$ ($T = \pi$)

Final summary

- ▶ A **real function of a real variable** $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is visualized through its **graph**
- ▶ The properties are stated on a subset $I \subseteq D$ of the domain; with no indication, $I = D$ is understood
- ▶ **Sign**: nonnegative, positive, nonpositive, negative functions in I
- ▶ **Boundedness**: the graph is contained in a half-plane or in a horizontal strip (upper and lower bounds)
- ▶ **Symmetries**: even ($f(-x) = f(x)$) and odd ($f(-x) = -f(x)$) functions
- ▶ **Monotonicity**: non-decreasing, increasing, non-increasing, decreasing functions in I
- ▶ **Periodicity**: $f(x + T) = f(x)$, as for the trigonometric functions